Component Selection /Footprint Selection

• Power and Regulation:

o 5V Input: A standard 1x02 pin was selected to connect in the external 5V power supply.

o Voltage Regulator: An AMS1117-3.3 linear regulator in a SOT-223 package steps down the 5V input to a stable 3.3V to power the components.

• Decoupling and Filtering:

o Capacitors: Four standard surface-mount ceramic capacitors (C_0805_2012Metric) consisting of three 100nF capacitors and one 10uF capacitor placed near power pins to filter high-frequency noise and for bulk decoupling.

• User Interface and Indicators:

o LEDs & Resistors: Three surface-mount LEDs (LED_0805_2012Metric) each paired with a 330-ohm resistor(R_0805_2012Metric) to visually indicate states safely without damage to the LED.

o Input Controls: A push button (SW_Push_SPST_NO_Alps_SKRK) for manual input and a vertical potentiometer (R_Potentiometer) for analog input sensing.

o Audio: A buzzer (Buzzer_12x9.5RM7.6) for audible feedback.

• Interface and Communication ports:

o STM32 Interface: Two 1x20 pin sockets (PinSocket_1x20_P2.54mm_Vertical) to securely mount and connect the STM32 board to the pcb.

o Expansion Ports: Standard 1x04 pin sockets

(PinSocket_1x04_P2.54mm_Vertical) designated for external I2C and UART communication modules.